Branched Multi-Turn GRPO for Code-Policy Generation on a Single GPU

Abstract

We train a small instruction-tuned LLM to emit Python control policies for sandboxed RL environments using a critic-free, on-policy multi-turn RL scheme. The model interacts with the environment over a two-turn chat-template protocol: a scratchpad or rehearsal turn, then a committed `policy` body whose scenario reward supplies the only scalar signal. We replace flat trajectory-level credit assignment with a $G_1 \times G_2$ branched rollout: G_1 first-turn variants per prompt, G_2 second-turn continuations per variant, scored at the leaves. Per-turn Monte Carlo value estimates yield per-turn advantages broadcast across the tokens of each turn, in the spirit of VinePPO but at chat-turn rather than intra-completion granularity. To fit the resulting trajectories into a 24 GB GPU we exploit the additivity of the token-summed clipped surrogate to decompose the loss across turns, bounding peak LM-head gradient memory to max(turn 1, turn 2) instead of their sum. The contribution is a clean, small-budget specialization of standard ingredients (GRPO, VinePPO, ArCHer-style hierarchical credit assignment) to the chat-template multi-turn code-policy setting, plus the per-turn loss decomposition that makes it tractable on a single L4.

1 Introduction

Reinforcement-learning fine-tuning of LLMs has converged on a small set of recipes: PPO with a learned critic, GRPO and its DAPO/CISPO variants which replace the critic with a group-relative Monte Carlo baseline, and process-supervision pipelines that train an auxiliary reward model. Most of this work targets single-turn reasoning, where the trajectory is a prompt followed by one completion and the reward is observed at the end. Extending the same recipes to multi-turn agentic settings exposes a credit-assignment problem: a single trajectory-level advantage broadcast to every model token cannot distinguish a strong first-turn strategy followed by a poor execution from the converse.

We study this problem in the concrete setting of *code-policy generation*: the LLM emits a Python `def policy(state)` body that is then executed against environment scenarios in a sandboxed subprocess. The artifact is the code; the LLM is removed from the inner control loop at deployment. We use two chat turns. The first either drafts a candidate `policy` (whose actual scenario reward is read back as a tool message) or runs a `main` scratchpad whose stdout is fed back; the second commits the scored `policy`.

Our training scheme has two ingredients. First, we replace trajectory-level credit assignment with a $G_1 \times G_2$ branched rollout and per-turn MC value estimates — the macro-step analog of VinePPO's intra-completion branching, with hierarchical baselines that vary by turn-1 prefix. Second, to make the resulting two-turn trajectories trainable on a 24 GB L4, we observe that the token-summed clipped surrogate is additive over disjoint token subsets sharing the same global normalizer, and exploit this to forward and backward turn-1 and turn-2 independently. Peak LM-head gradient memory drops from $O(B \cdot T_{\text{full}} \cdot |V|)$ to $O(B \cdot \max(T_1, T_2) \cdot |V|)$ at gradient cost zero; the parameter gradients summed across the two passes equal the single-pass gradient exactly.

The combination is a natural specialization of standard mechanisms rather than a new algorithm. We do not claim algorithmic novelty; the contribution is a clean assembly that fits on a single small GPU and the loss-decomposition trick that makes it possible.

2 Related work

Critic-free LLM RL. GRPO [1] and its token-summed variants (DAPO [2], CISPO) replace PPO's learned critic with a group-relative MC baseline. We use the token-summed form because its additivity over token subsets is what makes our per-turn loss decomposition exact.

Branching for credit assignment. The closest existing method is VinePPO [3], which estimates $\widehat{V}(s_t)$ at chosen intra-completion states via on-policy MC re-branching and uses it as a per-token baseline. We use the same mechanism but branch only at *turn boundaries* of a chat trajectory and broadcast the resulting advantage as a constant within each turn — group-relative within a turn, VinePPO-like across turns.

Hierarchical multi-turn agents. ArCHer [6] adopts the same macro-step decomposition (per-turn value, per-token policy update) but learns the per-turn value with a critic. We estimate it by MC over G_2 continuations, trading variance for the cost of fitting a value head. ReBN [5] replaces GAE with batch-normalized MC returns; the hierarchical mean we use can be viewed as ReBN partitioned by turn-1 prefix. RAGEN's StarPO [4] is the canonical flat-baseline critic-free MC scheme for multi-turn LLM RL; setting $G_2 = 1$ collapses our scheme to it.

$G_1 \times G_2$ tree rollouts. The same tree shape appears in process-reward-model training pipelines such as Math-Shepherd [7] and OmegaPRM [8], where the per-prefix MC mean labels intermediate steps for a learned reward model. We use the tree as a policy-gradient baseline rather than as a labelling oracle.

Code policies. Voyager [9] and FunSearch [10] also produce executable code as the artifact, but rely on frozen-LLM prompting plus skill-library accumulation or evolutionary mutation respectively. We adapt the LLM by gradient-based RL of LoRA weights and add a scratchpad turn between prompt and committed code.

3 Method

3.1 Setup

The agent is a language model π_θ over chat-template trajectories. The base is `Qwen2.5-Coder-3B-Instruct` with frozen weights W; θ parameterizes a LoRA adapter $W' = W + AB^\top$. A task $\mathcal{T}$ specifies environment source code, S scenarios $\{\sigma_i\}$ that vary the seed and hidden parameters, and a per-scenario reward. The scoring oracle runs a candidate policy body in a sandboxed subprocess against each scenario and returns $R(\text{code}; \mathcal{T}) = \sum_i r_i(\text{code}; \sigma_i)$. Validation failures, errors, and timeouts on σ_i score the *error floor* $r_i = -200$, calibrated below any running-but-failing trajectory — otherwise errored rollouts ($r = 0$) outrank running-but-poor rollouts and the gradient rewards crashing.

3.2 Two-turn protocol ($N = 2$)

Each turn the model emits one fenced code block tagged `policy` or `main`. `policy` defines `def policy(state)`, which the tool executes across all scenarios and reports the summed reward as a user-role message. `main` is arbitrary scratchpad code with no environment access; the tool returns truncated stdout. The final turn must be `policy`; `main` or untagged final turns are

floored. Malformed intermediate turns are likewise floored and surfaced as a textual rejection. The full trajectory is $\tau = (s,\, y^{(1)},\, m,\, y^{(2)})$, with $y^{(1)}, y^{(2)}$ the two completions, m the user-role tool message after $y^{(1)}$, and $R(\tau) = R(\text{code in } y^{(2)})$ the only scalar reward.

3.3 Token-summed clipped surrogate

For trajectory b at position t, with importance ratio $\rho_t^b = \pi_\theta / q$ and per-token advantage A_t^b,

$$\mathcal{L}(\theta) = -\frac{1}{N_{\text{tok}}} \sum_{b,t} \mathbf{1}\{\text{model token}\} \cdot \min\big(\rho_t^b A_t^b,\, \text{clip}(\rho_t^b,\, 1-\epsilon_{\text{lo}},\, 1+\epsilon_{\text{hi}}) A_t^b\big), \tag{1}$$

with N_{tok} the count of model tokens in the global batch.[1] (1) is *additive over disjoint token subsets*: for any partition of model-token positions, the loss equals the sum of partial losses computed with the same N_{tok}. This is the property the per-turn decomposition (§3.6) exploits.

3.4 Branched rollout

For each prompt s, sample G_1 turn-1 variants in one `model.generate` call: $y_i^{(1)} \sim q(\cdot \mid s)$, $i = 1, \ldots, G_1$. For each $y_i^{(1)}$ execute the tool to obtain m_i, then sample G_2 continuations $y_{i,j}^{(2)} \sim q(\cdot \mid s, y_i^{(1)}, m_i)$ in one call. Score each leaf to obtain $\{R_{i,j}\}$. Generate-call cost per prompt is $1 + G_1$, versus $2G_1G_2$ for an unbranched rollout with the same number of leaves.

3.5 Per-turn MC value estimates and advantages

Two empirical values, both pure averages over the tree:

$$\widehat{V}_0 = \frac{1}{G_1 G_2} \sum_{i,j} R_{i,j}, \qquad \widehat{V}_1^{(i)} = \frac{1}{G_2} \sum_{j} R_{i,j}. \tag{2}$$

$\widehat{V}_0$ estimates the initial-state value; $\widehat{V}_1^{(i)}$ the value after committing to turn-1 variant $y_i^{(1)}$. Per-turn advantages, broadcast to every token of the corresponding turn:

$$A_i^{(1)} = \widehat{V}_1^{(i)} - \widehat{V}_0 \qquad \text{(over } y_i^{(1)}\text{)}, \tag{3}$$

$$A_{i,j}^{(2)} = R_{i,j} - \widehat{V}_1^{(i)} \qquad \text{(over } y_{i,j}^{(2)}\text{)}. \tag{4}$$

The turn-2 advantage uses the conditional mean given the turn-1 prefix rather than the global per-prompt mean: a poor turn-2 sample drawn from a strong turn-1 is recognized as poor instead of inheriting its prefix's above-average outcome. Tool-message tokens are masked via `tool_mask`; their advantage is zero.

3.6 Per-turn loss decomposition

A single forward over $[s \,|\, y^{(1)} \,|\, m \,|\, y^{(2)}]$ materializes an LM-head gradient of shape $(B,\, T_{\text{full}},\, |V|)$. With $|V| \approx 152{,}000$ this exceeds 24 GB at our trajectory lengths. Let $\mathcal{I}_1, \mathcal{I}_2$ index turn-1 and turn-2 model tokens. Since the tool mask zeroes the loss elsewhere and (1) sums over positions with a single normalizer,

$$\mathcal{L} = \mathcal{L}_{\mathcal{I}_1} + \mathcal{L}_{\mathcal{I}_2}, \qquad N_{\text{tok}} = |\mathcal{I}_1| + |\mathcal{I}_2| \text{ shared}. \tag{5}$$

We compute the two terms in separate forward/backward passes:

[1] This matches the DAPO and CISPO recipes; per-row averaging variants (original GRPO, BNPO, Dr.GRPO) divide by per-row token counts.

- **Pass 1.** Forward $[s \,|\, y^{(1)}]$ with `logits_to_keep`$= T_1^{\max}$, advantages $A_i^{(1)}$. Compute $\mathcal{L}_1$, backward, free.
- **Pass 2.** Forward $[s \,|\, y^{(1)} \,|\, m \,|\, y^{(2)}]$ but pack $[s \,|\, y^{(1)} \,|\, m]$ into the prompt slot and $y^{(2)}$ alone into the completion slot, so `logits_to_keep`$= T_2^{\max}$. Compute $\mathcal{L}_2$, backward, free.

By (5) the accumulated parameter gradients sum to the single-pass gradient exactly. Peak memory is max(pass 1, pass 2) since pass-1's graph is freed before pass-2's forward begins.

Restrictions. The decomposition requires the token-summed, single-normalizer form of (1); per-row averaging variants do not factor across $\mathcal{I}_1 \cup \mathcal{I}_2$. The implementation also needs $\beta = 0$ (a KL term to π_{ref} would require per-segment slicing) and `num_iterations`$= 1$ (so cached `old_per_token_logps` need no re-segmenting).

4 Experimental setup

Environment. A 2D puck-on-tilted-plane controller. The puck moves in $[-6, 6]^2$ under a constant hidden tilt (t_x, t_y) and linear drag μ; the policy receives $\{p_x, p_y, v_x, v_y, g_x, g_y\}$ and must drive the puck to a goal within 120 steps. Hidden parameters — tilt, drag, thrust limit $a_{\max}$, initial velocity — vary per scenario and are not in the state. The thrust envelope is set to $a_{\max} = \|\text{tilt}\|_2 \times 1.05 + \epsilon$ so a naive straight-at-goal policy fails because it cannot cancel tilt drift. Per-step reward is $-\Delta t \cdot \text{dist}(p, g)$ with a $+50$ bonus on reaching the goal; errored episodes score -200 per scenario.

Baseline gap. Two hand-written policies fix the headroom: naive aim ($\approx 26\%$ goal rate over 200 seeds) and PD-with-tilt-feedforward ($\approx 73\%$). The gap is the room LoRA fine-tuning has to exploit.

Model and compute. `Qwen2.5-Coder-3B-Instruct`, LoRA rank $r = 16$ on attention and MLP projections, on a single 24 GB L4 GPU. bf16 weights, gradient checkpointing on. We use $G_1 = G_2 = 4$ (16 leaves per prompt) and a per-turn token cap of 400.

Training. The token-summed clipped surrogate (1) with $\epsilon_{\text{lo}} = 0.2$, asymmetric upper clip, $\beta = 0$, and the per-turn advantages of (3)–(4). No reference KL, no critic, no GAE, no context truncation across turns.

Diagnostics. We log $\widehat{V}_0$ and the between-variant standard deviation of $\widehat{V}_1$, the within-variant standard deviation, and their ratio (the *signal ratio*); the per-turn advantage standard deviations; the fraction of trajectories at the error floor; and the count of `generate` calls per prompt (expected $1 + G_1$). The signal ratio guides allocation between G_1 and G_2 at fixed budget: > 1 means turn-1 variation dominates and a larger G_1 helps; < 1 means within-variant noise dominates and a larger G_2 helps.

5 Evaluating the per-turn credit assignment

The per-turn loss decomposition (§3.6) is purely a memory enabler — it does not change the gradient. The substantive claim that needs empirical support is the *credit-assignment* contribution: that hierarchical per-turn advantages (3)–(4) on a $G_1 \times G_2$ tree do better than a single trajectory-level advantage broadcast across both turns. We propose a four-arm ablation (Table 1) that isolates each ingredient under a matched compute budget.

Arm	Rollout structure	Advantage	What it isolates
A — Flat StarPO	G independent N=2 trajectories	$A(\tau) = R(\tau) - \bar{R}$, broadcast to all tokens	Vanilla multi-turn GRPO baseline (RAGEN-style)
B — Branched, flat	$G_1 \times G_2$ tree	$A_{i,j} = R_{i,j} - \bar{R}$ on all tokens	Does the tree alone help (more leaves per generate call), independent of credit?
C — Branched, hierarchical (ours)	$G_1 \times G_2$ tree	$A_i^{(1)} = \widehat{V}_1^{(i)} - \widehat{V}_0$ on $y_i^{(1)}$; $A_{i,j}^{(2)} = R_{i,j} - \widehat{V}_1^{(i)}$ on $y_{i,j}^{(2)}$	The per-turn baseline contribution
D — Single-turn GRPO	G flat N=1 rollouts	Standard GRPO advantage	Does multi-turn help at all?

Table 1: Ablation arms. Match leaf count $G = G_1 G_2 = 16$ across A, B, C so they see the same number of scored trajectories per step. A uses $2G$ `generate` calls per prompt; C uses $1 + G_1$. Report both equal-rollouts and equal-wall-clock comparisons.

Primary metric. Held-out goal rate on a stress distribution (more extreme tilts/drag than train) at the best checkpoint, reported as median and IQR over ≥ 3 seeds per arm, plotted against gradient steps and against wall-clock. Statistical test: paired Wilcoxon over per-seed best-checkpoint goal rates; report effect size, not just p. The minimum convincing comparison is C > A at matched leaves.

Diagnostic checks. Two further measurements sharpen the mechanism story.

1. *Turn-1 quality curve.* Track the mean reward of turn-1 `policy` drafts (when turn-1 is in policy mode) as a function of training step. Under A the trajectory-level advantage projects onto turn-1 only through the global mean; under C, $A_i^{(1)}$ provides a direct gradient signal on turn-1 quality. C should show a steeper turn-1 reward rise than A.
2. *Signal-ratio conditioning.* The training loop already logs the per-prompt signal ratio $\widehat{\sigma}(\widehat{V}_1^{(i)})/\overline{\widehat{\sigma}(R_{i,\cdot})}$. Bucket prompts by their iteration-0 signal ratio. Ratio > 1 means turn-1 choice dominates the variance and per-turn credit should help; ratio < 1 means within-variant noise dominates and per-turn credit should not. C's advantage over A should grow with the bucket's signal ratio. If it does not, the mechanism is not doing what theory predicts and the headline claim weakens.

Failure modes to declare upfront. If C $\not>$ A under matched leaves, the hierarchical baseline does not pay off in this domain and the contribution reduces to "branching amortizes generate calls" (B vs A). If C $\approx$ B but both beat A, the gain is from the tree structure, not from the per-turn baseline. Either outcome is informative; we report whichever obtains.

6 Limitations

The scheme is restricted to $N = 2$ turns. Generalizing to $N > 2$ requires a recursive value bootstrap (each $\widehat{V}_n^{(\cdot)}$ becomes the conditional mean over the subtree below it), and the two-pass loss decomposition becomes an N-pass decomposition; neither is conceptually new but both add bookkeeping. We assume deterministic environments given the seed — under stochastic dynamics the leaf rewards become noisier estimates of state values and G_2 would have to grow accordingly. The $\beta = 0$, `num_iterations`$= 1$ restrictions are implementation choices that the same decomposition argument can lift with extra per-segment slicing of reference log-probabilities and cached sampling log-probabilities.

7 Conclusion

We described a critic-free, on-policy multi-turn RL scheme for code-policy generation that combines branched rollouts with per-turn MC value estimates and a per-turn loss decomposition. The algorithmic ingredients are individually standard; the assembly is small enough to train on a single 24 GB GPU, and the loss-decomposition trick is what makes the assembly tractable.

References

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